

1. Considere la sucesión definida recursivamente por:

$$\begin{cases} y_0 = 1 \\ y_n = \frac{1}{2}y_{n-1} + 3, \quad n \geq 1 \end{cases}$$

Entonces

a) Calcule y_3, y_4 .

(2 puntos)

b) Deduzca una fórmula explícita para y_n y compruebe que se puede escribir como:

$$y_n = \frac{1}{2^n} + 6 \left(1 - \frac{1}{2^n} \right)$$

(Sug. use la fórmula para $\sum_{k=0}^n r^k$ que demostró en la Tarea 1).

(6 puntos)

c) Calcule $\lim_{n \rightarrow \infty} y_n$

(2 puntos)

$$\begin{aligned} & \swarrow n=0 \\ y_0 &= 1 \end{aligned}$$

$$\swarrow n=1$$

$$y_1 = \frac{1}{2} \cdot 1 + 3$$

$$y_2 = \frac{1}{2} \left(\frac{1}{2} \cdot 1 + 3 \right) + 3$$

$$y_3 = \frac{1}{2} \cdot \left[\frac{1}{2} \left(\frac{1}{2} \cdot 1 + 3 \right) + 3 \right] + 3$$

$$y_4 = \frac{1}{2} \cdot \left[\frac{1}{2} \cdot \left[\frac{1}{2} \left(\frac{1}{2} \cdot 1 + 3 \right) + 3 \right] + 3 \right] + 3$$

$$y_n = \frac{1}{2^n} + 6 \left(1 - \frac{1}{2^n} \right) \quad \begin{cases} y_0 = 1 \\ y_n = \frac{1}{2}y_{n-1} + 3, \quad n \geq 1 \end{cases}$$

$$n=1 \quad \frac{1}{2^1} + 6 \left(1 - \frac{1}{2^1} \right) = \frac{1}{2} \cdot 1 + 3$$

$$\frac{1}{2} = \frac{1}{2} \quad \checkmark$$

$$y_n = \frac{1}{2^n} + 6 \left(1 - \frac{1}{2^n}\right) \begin{cases} y_0 = 1 \\ y_n = \frac{1}{2}y_{n-1} + 3, n \geq 1 \end{cases}$$

$$h=1 \quad y_p = \frac{1}{2^1} + 6 \left(1 - \frac{1}{2^1}\right) \quad \text{H.I}$$

$$h=p+1 \quad y_{p+1} = \frac{1}{2^{p+1}} + 6 \left(1 - \frac{1}{2^{p+1}}\right) \rightarrow \frac{1}{2^{p+1}} + 6 - \frac{6}{2^{p+1}}$$

$$y_{p+1} = \frac{1}{2} y_{(p+1)-1} + 3$$

$$\frac{1 - 6}{2^{p+1}} + 6$$

$$= \frac{1}{2} \cdot y_p + 3$$

Hi

$$6 - \frac{5}{2^{p+1}}$$

$$\frac{1}{2^1} \left[\frac{1}{2^1} + 6 \left(1 - \frac{1}{2^1}\right) \right] + 3$$

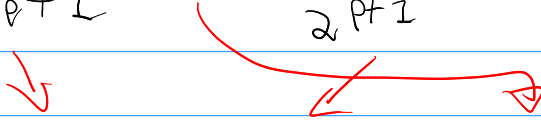
$$\frac{1}{2^{p+1}} + 3 \left(1 - \frac{1}{2^p}\right) + 3$$

$$\frac{1}{2^{p+1}} + 3 - \frac{3}{2^p} + 3$$

$$\frac{1}{2^{p+1}} + 3 - \frac{3}{2^p} \cdot \frac{2}{2} + 3$$

$$\frac{1}{2^{p+1}} + 3 - \frac{6}{2^{p+1}} + 3$$

$$\frac{1}{2^{p+2}} + 3 - \frac{6}{2^{p+1}} + 3$$

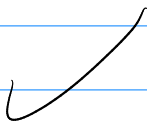
$$\frac{1}{2^{p+2}} + 6 - \frac{6}{2^{p+1}}$$


$$\frac{1}{2^{p+2}} - \frac{6}{2^{p+1}} + 6$$

$$\frac{1-6}{2^{p+1}} + 6$$

$$6 - \frac{5}{2^{p+1}}$$

$$\frac{1}{2^{p+1}} + 6 \left(1 - \frac{1}{2^{p+1}} \right)$$



$$\lim_{n \rightarrow +\infty} \frac{1}{2^n} + 6 \left(1 - \frac{1}{2^n} \right)$$

$$\frac{\cancel{1}}{\cancel{+\infty}} + 6 \left(1 - \frac{\cancel{1}}{\cancel{+\infty}} \right)$$

$$\boxed{6}$$

